

Chapter 1

The Main Problem of Statistics

Statistics begins with a question that is deceptively simple: *what can we know about a large group by studying a smaller part of it?* A doctor cannot test every human to measure average cholesterol. An election analyst cannot ask every voter how they will vote. A manufacturer cannot inspect every part off the assembly line. In each case, we observe a *sample* and use it to draw conclusions about the *population*.

Everything in this course — every formula, every test, every graph — is a tool for doing this carefully. This chapter introduces the vocabulary, the notation, and the first numerical summaries that make the rest of statistics possible.

1.1 Two Worlds: Population and Sample

Before computing anything, we must understand what we are trying to describe.

Definition: Population. The **population** is the complete collection of all individuals or objects under study. It is the group we *want* to describe. Populations are usually too large to observe directly.

Definition: Sample. A **sample** is a subset of the population that we actually observe and measure. We use the sample to estimate properties of the population.

Definition: Parameter. A **parameter** is a numerical fact about the population. Parameters are usually *unknown* and are what we are trying to estimate. We denote parameters with **Greek letters**.

Definition: Statistic. A **statistic** is a numerical value computed from sample data. Statistics are *known* — we calculate them from our observations. We denote statistics with **Roman (English) letters**.

1.1.1 The Greek/Roman Distinction

This distinction is fundamental and appears throughout the entire course. A convenient reference:

Quantity	Population (parameter)	Sample (statistic)
Mean	μ (“mu”)	$\bar{x}$ (“x-bar”)
Standard deviation	σ (“sigma”)	s
Proportion	p	$\hat{p}$ (“p-hat”)
Size	N	n

Think of it this way: Greek letters are for the world we cannot fully see. Roman letters are for the world we measured ourselves.

Example 1.1.1. (*Population vs. Sample*) A college wants to know the average GPA of all its students (about 4,200 enrolled). A researcher surveys 150 randomly chosen students and finds their average GPA is 2.87.

- Population: all 4,200 enrolled students.
- Sample: the 150 surveyed students.
- Parameter: μ , the true average GPA of all 4,200 (unknown).
- Statistic: $\bar{x} = 2.87$, the average GPA of the 150 surveyed (known).

The researcher uses $\bar{x} = 2.87$ to *estimate* μ .

Common Pitfall. Students often use “population” to mean “all people in the world.” In statistics, the population is whatever complete group your study is *about*. If you are studying the eating habits of students at your college, the population is those students — not all humans, not all college students everywhere.

Exercises 1.1

- 1.1.1. A polling company calls 800 registered voters in Washington State and asks about their approval of the governor. Identify the (a) population, (b) sample, (c) parameter of interest, (d) statistic computed.

1.1.2. A veterinarian measures the weight of 30 cats brought in for annual checkups and finds the average is 10.4 pounds. She wants to know the average weight of all domestic cats in her city. Identify the four quantities above.

1.1.3. True or False: A parameter changes every time a new sample is taken. Explain your answer.

1.2 Types of Data

Not all data are numbers, and even among numbers, some behave differently.

Definition: Categorical Data. **Categorical** (or **qualitative**) data places each individual into a group or category. Examples: eye color, political party, type of car, yes/no responses.

Definition: Quantitative Data. **Quantitative** data are numerical measurements where arithmetic makes sense. There are two kinds:

- **Discrete:** countable values, often whole numbers. Examples: number of children, number of defective parts, number of customers per day.
- **Continuous:** values along an unbroken scale. Examples: height, temperature, time, weight.

Why does this matter? The type of data determines which summaries and tests are appropriate. You cannot compute the “average eye color.” You can compute the average height.

Example 1.2.1. (Classifying Variables) A survey asks students: (1) their major, (2) their GPA, (3) how many credit hours they are taking, (4) how many hours per week they study.

- Major: categorical.
- GPA: continuous quantitative (in principle, any decimal value in $[0.0, 4.0]$).
- Credit hours: discrete quantitative (whole numbers: 3, 6, 12, ...).
- Study hours: continuous quantitative (though often reported as a whole number).

Exercises 1.2

- 1.2.1. Classify each variable as categorical, discrete quantitative, or continuous quantitative: (a) number of cars in a household; (b) annual income; (c) brand of coffee preferred; (d) blood type; (e) distance driven to campus; (f) score on a 10-question quiz.
- 1.2.2. Give one example each of a categorical variable and a continuous variable that you could measure about the students in this class.

1.3 Summation Notation

Before we can write formulas for the mean and standard deviation, we need a compact way to write “add up all the values.” Mathematicians use the **summation symbol** Σ , which is the capital Greek letter *sigma*.

1.3.1 Reading the Sigma Symbol

The expression

$$\sum_{i=1}^n x_i$$

is read: “the sum of x_i from $i = 1$ to n .”

Here is what each piece means:

- x_i is the i -th data value. The subscript i is an index that counts the values.
- $i = 1$ below the Σ is the starting index.
- n above the Σ is the ending index.
- The whole expression says: replace i with $1, 2, 3, \dots, n$ in turn, and add up all the results.

Example 1.3.1. (*Expanding Summation Notation*) Suppose we have five data values: $x_1 = 3$, $x_2 = 7$, $x_3 = 2$, $x_4 = 9$, $x_5 = 5$. Then:

$$\sum_{i=1}^5 x_i = x_1 + x_2 + x_3 + x_4 + x_5 = 3 + 7 + 2 + 9 + 5 = 26.$$

Example 1.3.2. (*Summation with Squaring*) Using the same five values, compute

$$\sum_{i=1}^5 x_i^2.$$

We square *each* value first, then add:

$$\sum_{i=1}^5 x_i^2 = 3^2 + 7^2 + 2^2 + 9^2 + 5^2 = 9 + 49 + 4 + 81 + 25 = 168.$$

Important: $\sum x_i^2 \neq \left(\sum x_i\right)^2$. We have $\sum x_i^2 = 168$ but $\left(\sum x_i\right)^2 = 26^2 = 676$. These are completely different.

Common Pitfall. $\sum x_i^2$ and $\left(\sum x_i\right)^2$ are not equal. The first squares each value, then adds. The second adds first, then squares. Order matters.

In practice, once the meaning is clear, we often drop the explicit indices and write $\sum x$ as shorthand for $\sum_{i=1}^n x_i$. We will use both notations in this course.

Exercises 1.3

1.3.1. Given $x_1 = 4$, $x_2 = 1$, $x_3 = 6$, $x_4 = 3$, compute: (a) $\sum_{i=1}^4 x_i$; (b) $\sum_{i=1}^4 x_i^2$; (c) $\left(\sum_{i=1}^4 x_i\right)^2$.

1.3.2. If $\sum_{i=1}^6 x_i = 42$, what is $\sum_{i=1}^6 (x_i + 3)$? (*Hint*: distribute the sum.)

1.3.3. Write out in plain English what $\sum_{i=1}^n (x_i - \bar{x})$ means. (Don't compute anything — just describe the process in words.)

1.4 Frequency Tables and Relative Frequency

When we have a dataset, our first goal is usually to understand its shape — where values cluster, where they spread, whether there are outliers.

Definition: Frequency. The **frequency** of a value or class is the count of how many times it appears in the data.

Definition: Relative Frequency. The **relative frequency** of a value or class is its frequency divided by the total number of observations:

$$\text{relative frequency} = \frac{\text{frequency}}{n}.$$

Relative frequencies always sum to 1 (or equivalently, 100%).

Relative frequency is more useful than raw frequency when comparing groups of different sizes.

Example 1.4.1. (*Building a Frequency Table*) Twenty students report the number of hours of sleep they got last night:

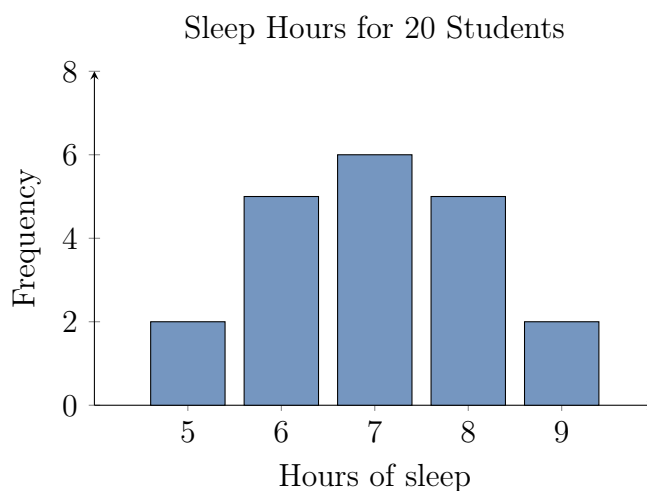
6, 7, 8, 6, 5, 7, 8, 9, 6, 7, 7, 8, 6, 5, 9, 7, 8, 6, 7, 8.

Hours of sleep	Frequency	Relative frequency
5	2	$2/20 = 0.10$
6	5	$5/20 = 0.25$
7	6	$6/20 = 0.30$
8	5	$5/20 = 0.25$
9	2	$2/20 = 0.10$
Total	20	1.00

Seven hours was the most common response, accounting for 30% of students.

1.4.1 Histograms

For continuous data (or discrete data with many values), we group observations into **classes** (also called *bins*) and draw a histogram.



Exercises 1.4

- 1.4.1. Fifteen quiz scores are recorded: 70, 82, 91, 75, 88, 70, 95, 82, 77, 91, 70, 88, 82, 75, 91. Construct a frequency and relative frequency table.
- 1.4.2. In the table you built, which score has the highest relative frequency? What percentage of students earned that score?
- 1.4.3. If a second class of 30 students took the same quiz, explain why comparing relative frequencies is more informative than comparing raw frequencies.

1.5 Measures of Center: Mean and Median

A single number that summarizes “where a dataset is centered” is called a **measure of center**. The two most important are the mean and the median.

1.5.1 The Sample Mean

Definition: Sample Mean. The **sample mean** $\bar{x}$ (read “x-bar”) is the arithmetic average of the sample values:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

The corresponding population parameter is μ (“mu”).

Example 1.5.1. (*Computing the Sample Mean*) Seven students’ exam scores: 72, 85, 90, 68, 95, 78, 83.

$$\bar{x} = \frac{72 + 85 + 90 + 68 + 95 + 78 + 83}{7} = \frac{571}{7} \approx 81.57.$$

The average score was about 81.6 points.

1.5.2 The Median

Definition: Median. The **median** is the middle value when data are sorted in order.

- If n is odd, the median is the value at position $\frac{n+1}{2}$.

- If n is even, the median is the average of the values at positions $\frac{n}{2}$ and $\frac{n}{2} + 1$.

Example 1.5.2. (*Computing the Median — Odd n*) The seven scores from Example 1.5.1 sorted: 68, 72, 78, 83, 85, 90, 95.

There are $n = 7$ values, so the median is at position $(7 + 1)/2 = 4$. The 4th value is **83**.

Example 1.5.3. (*Computing the Median — Even n*) Four prices: \$12, \$18, \$24, \$31. $n = 4$, so the median is the average of the 2nd and 3rd values:

$$\text{median} = \frac{18 + 24}{2} = 21.$$

1.5.3 Mean vs. Median: When to Use Each

The mean uses every data value, so it is sensitive to extreme values (outliers). The median is not affected by outliers.

Example 1.5.4. (*Effect of an Outlier*) Annual salaries at a small company: \$42k, \$45k, \$47k, \$49k, \$51k, \$250k (the owner).

Mean: $(\$42 + \$45 + \$47 + \$49 + \$51 + \$250)\text{k}/6 = \$484\text{k}/6 \approx \80.7k .

Median: average of 3rd and 4th values = $(\$47 + \$49)\text{k}/2 = \$48\text{k}$.

Which better describes a “typical” employee’s salary? The median, at \$48k, is far more representative than the mean at \$80.7k, which is inflated by the owner’s salary.

Common Pitfall. When a distribution is skewed right (a few very large values), the mean is pulled above the median. In such cases — household incomes, housing prices, waiting times — the median is usually the more meaningful measure of center.

Exercises 1.5

1.5.1. Compute the mean and median of: 14, 22, 9, 31, 17, 22, 8.

1.5.2. A dataset has mean 50. A new observation of 200 is added. Without computing, does the new mean increase, decrease, or stay the same? By a lot or a little? Explain.

1.5.3. House prices in a neighborhood: \$180k, \$195k, \$205k, \$212k, \$950k. Compute mean and median. Which is a better summary of a “typical” price? Why?

1.6 Measures of Spread: Standard Deviation

Two datasets can have the same mean but be very different. Consider:

Dataset A: 50, 50, 50, 50, 50. Mean = 50.

Dataset B: 10, 30, 50, 70, 90. Mean = 50.

Dataset B is far more spread out. We need a number to capture this.

1.6.1 Deviation from the Mean

The **deviation** of a value x_i is how far it is from the mean:

$$x_i - \bar{x}.$$

Deviations are positive for values above the mean, negative for values below. Notice:

$\sum_{i=1}^n (x_i - \bar{x}) = 0$ always. The deviations cancel out. To measure total spread, we cannot just sum the deviations — they will always cancel.

1.6.2 The Sample Standard Deviation

We fix the cancellation problem by squaring each deviation before summing:

Key Formula. The **sample standard deviation** s is:

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}.$$

The expression inside the square root, $\frac{\sum (x_i - \bar{x})^2}{n - 1}$, is called the **sample variance**, denoted s^2 .

The corresponding population parameter is σ (“sigma”) for standard deviation and σ^2 for variance. The population formula uses N in the denominator instead of $n - 1$.

Why $n - 1$ instead of n ? Using $n - 1$ corrects for a subtle bias: when we estimate μ with $\bar{x}$, we use up one degree of freedom. Dividing by $n - 1$ makes s^2 an unbiased estimator of σ^2 . This will matter in later chapters.

Example 1.6.1. (*Computing Standard Deviation by Hand*) Data: 3, 7, 7, 2, 6. $n = 5$.

Step 1: Compute $\bar{x}$.

$$\bar{x} = \frac{3 + 7 + 7 + 2 + 6}{5} = \frac{25}{5} = 5.$$

Step 2: Compute each deviation and its square.

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
3	-2	4
7	+2	4
7	+2	4
2	-3	9
6	+1	1
Sum = 0		Sum = 22

Step 3: Compute s .

$$s = \sqrt{\frac{22}{5-1}} = \sqrt{\frac{22}{4}} = \sqrt{5.5} \approx 2.35.$$

The data values are, on average, about 2.35 units away from the mean of 5.

Common Pitfall. The formula uses $n - 1$ (sample standard deviation). Many calculators have both a “ σ ” key (uses n , for populations) and an “ s ” key (uses $n - 1$, for samples). In this course, you will almost always want the s key.

1.6.3 Interpreting Standard Deviation

Standard deviation has the same units as the original data (because we took a square root at the end). A small s means data are tightly clustered around the mean; a large s means they are widely spread. There is no universal threshold for “small” or “large” — it always depends on context.

Exercises 1.6

1.6.1. Compute the sample standard deviation of: 10, 14, 12, 16, 13. Show the deviation table.

1.6.2. Two classes take the same exam. Class 1: mean 75, $s = 3$. Class 2: mean 75, $s = 18$. Describe in words what these numbers tell you about the two classes.

1.6.3. Without computing, decide which dataset has a larger standard deviation. Explain your reasoning.

Dataset A: 100, 101, 99, 100, 100.

Dataset B: 80, 90, 100, 110, 120.

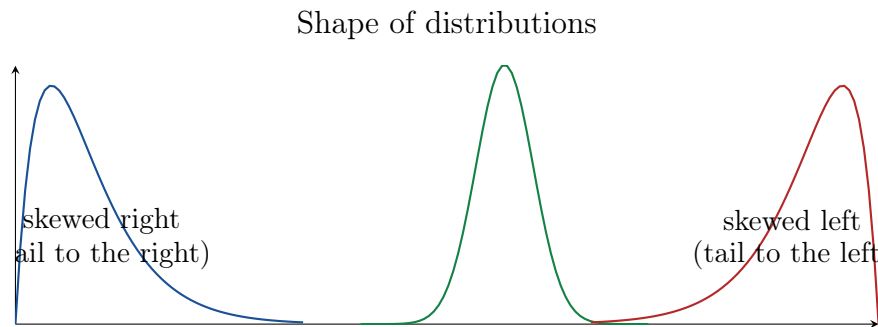
1.7 The Shape of Data

Beyond center and spread, we want to describe the **shape** of a distribution.

Definition: Symmetric Distribution. A distribution is **symmetric** if the left and right halves are mirror images. In a symmetric distribution, $\text{mean} \approx \text{median}$.

Definition: Skewed Distribution. A distribution is **skewed right** (positively skewed) if it has a long tail extending to the right. In this case, $\text{mean} > \text{median}$. A distribution is **skewed left** (negatively skewed) if the tail extends to the left. In this case, $\text{mean} < \text{median}$.

The direction of skew is the direction of the tail, not the bulk of the data. A useful mnemonic: the mean is pulled toward the tail.



Exercises 1.7

- 1.7.1. Ages of patients at an emergency room tend to include many young adults and some very elderly patients. Would you expect this distribution to be skewed left, skewed right, or symmetric? Explain.
- 1.7.2. For a dataset where $\text{mean} = 45$ and $\text{median} = 52$, what does this tell you about the shape of the distribution?

Chapter 1 Review

1. A researcher surveys 200 of the 5,000 employees at a company to estimate the average commute time. Identify (a) population, (b) sample, (c) parameter, (d) statistic. Use correct notation.

2. The following data are the ages of students in a statistics class:
18, 19, 21, 20, 22, 19, 18, 25, 20, 21, 18, 19, 30, 20, 21.
 - (a) Build a frequency and relative frequency table.
 - (b) Compute $\bar{x}$ and the median. Are they close? Why or why not?
 - (c) Compute s . Show the deviation table.
 - (d) Describe the shape of this distribution.
3. Given $x_1 = 2$, $x_2 = 5$, $x_3 = 3$, $x_4 = 8$, $x_5 = 2$, compute:
(a) $\sum x_i$ (b) $\sum x_i^2$ (c) $(\sum x_i)^2$ (d) $\bar{x}$ (e) s .
4. *Writing prompt:* In two to three sentences, explain in your own words what a standard deviation measures and why we divide by $n - 1$ instead of n when working with a sample.